

Problem

The current through an initially uncharged 1-F capacitor is $\delta(t)$ A. The voltage across the capacitor is:

- (a) $u(t)$ V
- (b) $-1/2 + u(t)$ V
- (c) $e^{-t}u(t)$ V
- (d) $\delta(t)$ V

Step-by-step solution

Step 1 of 1

Consider an initially uncharged 1 F capacitor.

$$C = 1 \text{ F}$$

The current through the capacitor is $\delta(t)$.

$$i_c(t) = \delta(t)$$

The following is the expression for the voltage across the capacitor.

$$v_c(t) = \frac{1}{C} \int i(t) dt$$

Substitute $\delta(t)$ for $i_c(t)$, and 1 F for C .

$$\begin{aligned} v_c(t) &= \left(\frac{1}{1 \text{ F}} \right) \int \delta(t) dt \\ &= (1)u(t) \\ &= u(t) \text{ V} \end{aligned}$$

Thus, the voltage across the capacitor, $v_c(t)$ is $u(t)$ V and the correct choice is **a**.